

MUHAMMAD ADEEL

AI/ML Engineer | Generative AI | LLMs | RAG | AI Agents

Pakistan | +92 309 5408984 | muhammadadeel0a1@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

AI/ML engineer building systems across machine learning, deep learning, and generative AI — from transfer-learning computer vision models to retrieval-augmented LLM applications and tool-calling AI agents. Experienced across the ML lifecycle, including data preprocessing, model training and evaluation, and API-based deployment with FastAPI, Flask, and Docker. Works fluently in Python, PyTorch/TensorFlow, and LangChain/LangGraph.

TECHNICAL SKILLS

Programming & Data: Python, SQL, MySQL, Pandas, NumPy

Machine Learning: Scikit-learn, XGBoost, LightGBM, CatBoost, Regression, Classification, Ensemble Learning, Feature Engineering, Cross-Validation

Deep Learning & NLP: PyTorch, TensorFlow, Transformers, NLP, CNN, RNN, GNN, Transfer Learning

Generative AI & LLMs: LLMs, RAG, Gemini API, Embeddings, Vector Search, Reranking, Semantic Search, Prompt Engineering, LangChain, LangGraph, AI Agents, Tool Calling

Computer Vision: EfficientNetV2, DenseNet121, YOLO, BLIP, OCR, Image Classification, Object Detection, Image Captioning

Backend & Deployment: FastAPI, Flask, REST APIs, PostgreSQL, Docker, Git, GitHub

PROFESSIONAL EXPERIENCE

Data Analyst — Positive Solutions, Wah Cantt

Apr 2026 – Jul 2026

- Analyzed business datasets using Excel, MySQL, and Python to support recurring cross-department reporting
- Built and maintained dashboards to track key performance metrics and surface trends for stakeholders
- Wrote and optimized SQL queries (joins, aggregations, subqueries) to extract and clean data, improving reporting accuracy
- Automated repetitive data-processing tasks with Python, reducing manual effort and speeding up decision-making

Artificial Intelligence Trainee — NAVTTC, Islamabad

Oct 2025 – Mar 2026

- Trained in core AI/ML concepts, assisting in developing and testing models using Python, NumPy, and Pandas for data preprocessing and evaluation
- Explored generative AI fundamentals, including LLMs and prompt engineering, alongside hands-on ML/DL model experimentation

PROJECTS

Enterprise AI Knowledge Assistant — RAG / LLM [\[GitHub\]](#)

- Engineered a production-style RAG pipeline — document ingestion, chunking, embedding generation, vector retrieval and reranking — to ground LLM responses in enterprise document data
- Integrated the Gemini API as the generation layer with context-construction logic; deployed as a FastAPI service in a Dockerized architecture, with an evaluation process to measure retrieval and response quality

AI Customer Support Agent — LLM / AI Agents [\[GitHub\]](#)

- Designed a LangGraph-orchestrated AI agent that autonomously handles support workflows — intent understanding, knowledge retrieval, database lookups, and order actions — via multi-step tool calling instead of scripted flows
- Built a RAG-backed knowledge layer integrated with a PostgreSQL customer/order system, exposed via a FastAPI backend and containerized with Docker for reproducible deployment
- Automated end-to-end resolution of repetitive support queries, from natural language request to completed action, reducing dependency on manual responses

AI-Powered “Third Eye” for Visually Impaired Persons — Multimodal AI / Computer Vision [\[GitHub\]](#)

- Built a real-time multi-module assistive system combining DenseNet121 (currency classification), YOLO11n (object detection), BLIP (scene captioning), OCR, and text-to-speech, achieving 90% classification accuracy on a Pakistan-specific dataset

Spine Disease Detection System — EfficientNetV2 / Final Year Project [\[GitHub\]](#)

- Developed a computer vision system to classify spinal disorders (neural foraminal narrowing, subarticular stenosis, spinal canal stenosis) using the RSNA 2024 Lumbar Spine dataset
- Fine-tuned EfficientNetV2 with transfer learning, data augmentation, and sampling techniques to address class imbalance; deployed via Flask, achieving 95% classification accuracy

Heart Disease Prediction — Ensemble Machine Learning [\[GitHub\]](#)

- Built and compared classification models (Logistic Regression, Random Forest, SVM, XGBoost, LightGBM, CatBoost) on the Kaggle Playground Series S6E2 dataset, reaching an ensemble AUC of ~0.956 via probability averaging

EDUCATION

Bachelor of Computer Science — University of Haripur

Sep 2021 – Jun 2025

CERTIFICATIONS

- Machine Learning Specialization — Andrew Ng (Coursera) [\[Verify\]](#)
- Deep Learning Specialization — Andrew Ng (Coursera) [\[Verify\]](#)